

Middlesex College

The New Web



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Back in the early 80s and 2000s, the inception of the internet was a very different place compared to the endless communication network it is considered now. The internet was home to very singular functional static websites, mainly occupied by large organizations and government agencies. This also included a small minority of tech-savvy people who managed to create their own web pages through their ISP/WHP. This era was known as Web 1.0, and this style of web development continued until the development of HTML and CGI, which introduced dynamic and eventually client-end functionality.

As the mid-2010s approached, there was a shift in the development of web pages to our current system, Web 2.0. The focus of this system was to help create content, boost social networking, and develop collaborative tools. One of the main factors for this push was the navigation from closed-source technologies like Adobe Flash and Microsoft Silverlight to brand-new open-source tools such as JavaScript, HTML5, and resources like DOM manipulation. Eventually, this would propel the internet into the technology we have today: TikTok, YouTube, Online Shopping, and WebGL. The expansion of the internet also came with the expansion of internet-led companies to begin the collection of user data to create new technologies, track users, and develop targeted advertising on the web. One of the biggest negatives of company-stored data is that it is lost when deleted. A modern-day example of this would be a user getting banned from TikTok for breaking TOS and no longer being able to request access to their saved music, content, creators, logins, watch history, etcetera.

Web3, which shouldn't be mistaken for Web 3.0 (a completely different system), is a restructuring of the current web that focuses on decentralized data and users being responsible for their own data. Web3 also mentions a universal web-native currency (cryptocurrency) as the future of the internet.

Unfortunately, the idea of Web3 assumes that people are responsible enough to take care of their data and privacy. Security issues arise when talking about public accessibility and blockchain (digital collection of transactions) scalability and prominence. However, the biggest issue regarding Web3 would be scamming.

In the current day and age, phishing, malware, and keylogging are common cyber scams in the Web 2.0 world. The difference in Web3, however, is the availability of all sensitive information to a user. If a scammer with malicious intent wanted to steal data, persuading a user to send over private keys would be detrimental to their privacy, data, and possible safety—more damaging than most scams in 2.0.

Data Aspect	Current Model	Web3 Model
Data Ownership	Organizations own most user data through EULAs You can export data out but cannot import data	User data owned by individuals with a unique signature. User can choose to share private data to trusted credited sources, but that access can be revoked
Data Location	Organizations hold data within databases, and it is duplicated constantly through different apps.	Blockchain hosts all public data with verifiable creds. Private info is stored externally in an encrypted data hub.
Data Access	Data in applications can be sold, altered, removed, or even monetized without user knowledge or approval	Data on external hubs require authorization to access which can be revoked at any moment

The table above shows the differences between the current system of data storage, manipulation, ownership, and collection and the Web3 alternative.

As previously mentioned, personal security and internet literacy are top priorities in a Web3 environment. There are plenty of malicious threats that may pose a risk to the safety of individuals via certain top priority scams.

The first scam is blackmail, where a person threatens a user to release private or sensitive information unless they send the person compensation. This one is particularly high because of the leverage a scammer may have and whether they are bluffing or not. Either way, victims usually follow through with instructed orders.

The next scam is fraud, another dangerous situation where a person disguises themselves as a relative, celebrity, or less fortunate individual to receive compensation from a manipulated victim. Whether this involves real identity theft and transfer of personal data or promises to double money sent by victims, it is still prevalent.

The final high-priority scam is hacking. This is arguably the most dangerous because it involves unauthorized users entering one's account to directly deposit their funds into their own wallets. This is usually done without alerting the victim or the bank, which is why it makes it the hardest, yet most successful security threat.

A strong argument against the switch to Web3 is the burden and fear it would cause many individuals regarding their privacy in fear of being a victim to these exploits. This fear is also solidified with the current state of Web3 progress involving bugs and errors. Part of this problem is because of the amount anonymous to the blockchain system who may have a different set of ethics and regulations when designing the system. This also means that testing certain functions and creating fixes for them may lead to eventual exploitation from the people who originally tested them. Overall, as this project is still in the blueprinting phase, there are plenty of bugs that don't provide the most assurance to even the most thrilled Web3 enthusiasts.

All is not lost in terms of security in a Web3 system; there are ways to authenticate and mitigate potential dangers for the future of the web. Regarding fraud, a solution would be to have verification endpoints that ensure people are who they say they are online when requesting data. There is a term called Zero-Knowledge Proofs where users can verify authenticity without revealing too much personal information.

End-to-end encryption is of course mandatory to maintain privacy. Data leakage is a real and serious issue and is a very big and important factor in security. An article claims that over 61% of Americans have had some personal data breached before. Even the government has interfered with collecting data through phone providers like AT&T, specifically Project Hemisphere.

Digital education is crucial to establishing a long-lasting internet system. As things change, people stay in their old ways and refuse to adapt, but this is very dangerous. The primary target of call center scams are the elderly because predators know how vulnerable they are.

In terms of hacking, it will have to be an adaptable situation as the future of Web3 grows. Wherever there is a vulnerability or weak point, somebody will try and hack it. By building a strong system, it can be improved over time. It ensures that they are ready to fix the latest exploits. The overarching idea and concept of Web3 is strong, with a lot of passion revolving around returning data to the people. As long as cybersecurity can be managed internally and spread externally, it can be the future of the web as we know it.

Works Cited

Dimuro, Claudia. “61% of Americans Have Had Their Personal Data Breached.” *Governing*, Governing, 9 Feb. 2024, www.governing.com/management-and-administration/61-of-americans-have-had-their-personal-data-breached.

Wyden, Ron. “Wyden Hemisphere Surveillance Letter 11.20.23.” *United States Senate*, www.wyden.senate.gov/imo/media/doc/wyden_hemisphere_surveillance_letter_112023.pdf. Accessed 6 May 2025.

Yaga, Dylan. “A Security Perspective on the WEB3 Paradigm.” *NIST Internal Report*, nvlpubs.nist.gov/nistpubs/ir/2025/NIST.IR.8475.pdf. Accessed 6 May 2025.